

IT312.001 - Cybersecurity Foundations

Executive Summary

**Project #10: Registering HIDS Agents
with Wazuh SIEM in a Proxmox
Environment**

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Introduction

This project centered on the implementation of a comprehensive Host Intrusion Detection System (HIDS) pipeline within a Proxmox Virtual Environment. By integrating centralized telemetry collection and analysis, the exercise effectively simulated a professional Security Operations Center (SOC) workflow. The core objective involved deploying Wazuh agents across diverse operating systems, specifically utilizing both Ubuntu 22.04 and Kali Linux for the Linux-based installations, alongside a Windows Server instance. These agents were then registered with a centralized SIEM console to establish a unified monitoring framework. To conclude the implementation, the infrastructure's defensive efficacy was validated through controlled, simulated attacks launched from a Kali Linux environment.

Problem Statement

Enterprise environments are vulnerable to distributed threats such as brute-force attacks and unauthorized access. Detecting these requires a Master Collector database that gathers forwarded logs from Sensors or agents on individual machines. The challenge addressed was the correct installation and registration of these HIDS agents to ensure that attack traffic generated from a Kali Linux instance could be captured and analyzed centrally rather than remaining localized.

Methodology

1. Environment Provisioning (Proxmox):

- Deployed a target environment consisting of a Windows Server Main instance and an Ubuntu Server instance within the Proxmox hypervisor.

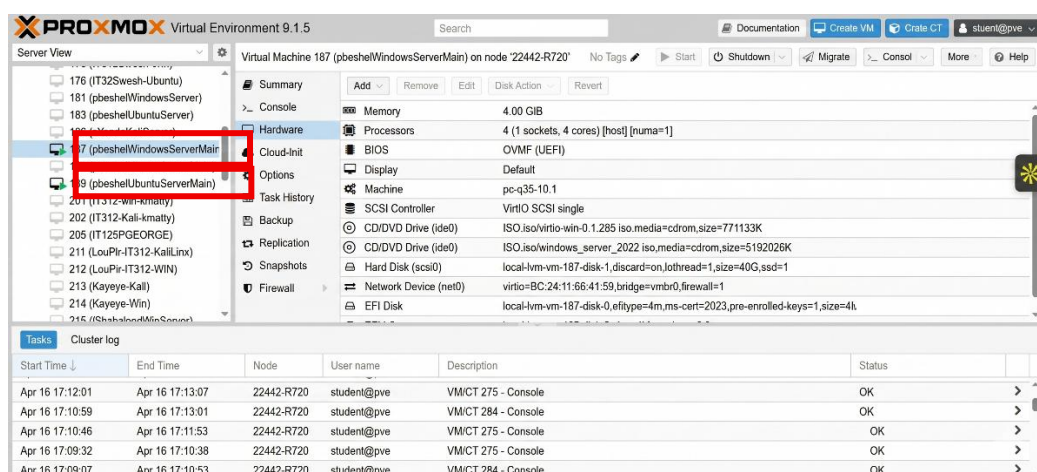


Fig.1 Proxmox Virtual Environment showing active Windows and Linux target VMs.

2. Manager Infrastructure Setup:

- Configured the Wazuh Manager on an Ubuntu host.
- Installed essential security and transport dependencies including *gnupg* and *apt-transport-https*.

```
root@wazuh-server:/home/wazuh# apt-get install gnupg apt-transport-https
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gnupg is already the newest version (2.2.27-3ubuntu2.4).
gnupg set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
0 upgraded, 1 newly installed, 0 to remove and 56 not upgraded.
Need to get 1,510 B of archives.
After this operation, 170 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 apt-transport-https all 2.4.14 [1,510 B]
Fetched 1,510 B in 1s (2,711 B/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 110646 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.4.14_all.deb ...
Unpacking apt-transport-https (2.4.14) ...
Setting up apt-transport-https (2.4.14) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuh-server:/home/wazuh#
```

Fig.2 Installing prerequisite dependencies (*gnupg*, *apt-transport-https*) on the Manager.

- Executed the *wazuh-install.sh* assistant to generate configuration files and certificates for the dashboard and indexer.

```

ubuntu@ubuntu:~$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh &
& sudo bash ./wazuh-install.sh -a
[sudo] password for ubuntu:
16/04/2026 14:22:29 INFO: Starting Wazuh installation assistant. Wazuh versi
on: 4.14.3
16/04/2026 14:22:29 INFO: Verbose logging redirected to /var/log/wazuh-insta
ll.log
16/04/2026 14:22:40 INFO: --- Dependencies ---
16/04/2026 14:22:40 INFO: Installing gawk.
16/04/2026 14:22:53 INFO: Verifying that your system meets the recommended m
inimum hardware requirements.
16/04/2026 14:22:53 INFO: Wazuh web interface port will be 443.
16/04/2026 14:23:06 INFO: --- Dependencies ---
16/04/2026 14:23:06 INFO: Installing apt-transport-https.
16/04/2026 14:23:12 INFO: Installing debhelper.
16/04/2026 14:24:15 INFO: Wazuh repository added.
16/04/2026 14:24:15 INFO: --- Configuration files ---
16/04/2026 14:24:15 INFO: Generating configuration files.
16/04/2026 14:24:15 INFO: Generating the root certificate.
16/04/2026 14:24:17 INFO: Generating Admin certificates.
16/04/2026 14:24:17 INFO: Generating Wazuh indexer certificates.
16/04/2026 14:24:17 INFO: Generating Filebeat certificates.
16/04/2026 14:24:18 INFO: Generating Wazuh dashboard certificates.
16/04/2026 14:24:19 INFO: Created wazuh-install-files.tar. It contains the W

```

Fig.3 Successful execution of the Wazuh installation assistant script.

3. HIDS Agent Deployment:

- **Windows:** Installed the agent via a remote MSI package using PowerShell, targeting the Manager IP 192.168.239.37.

```

PS C:\Users\Administrator> Invoke-WebRequest Uri "https://packages.wazuh.com/4.x/windows/wazuh-agents-4.14-1.msi" -OutFile
"C:\Windows\Temp\wazuh-agent.msi" -i $env:tmp wazuh-agent /q WAZUH_MANAGER="192.168.239.37" WAZUH_AGENT_NAME="Windows
"

```

Fig.4 PowerShell execution of the Wazuh Agent MSI installer targeting the Manager IP.

- **Linux:** Prepared the Ubuntu target by adding the Wazuh GPG key and repository to ensure a secure software supply chain.

```

ubuntu@ubuntu-Standard-PC-i440FX-PIIX-1996:~$ curl -s https://packages.wazuh.com
/key/PGP-KEY-WAZUH | sudo gpg --dearmor -o/usr/share/keyrings/wazuh-gpg
[sudo] password for ubuntu:

```

Fig.5 Adding the Wazuh GPG key on the Ubuntu target for secure repository access.

4. Registration & Verification:

- Confirmed the agents successfully moved to an Active status within the dashboard, proving the registration tunnel was open.

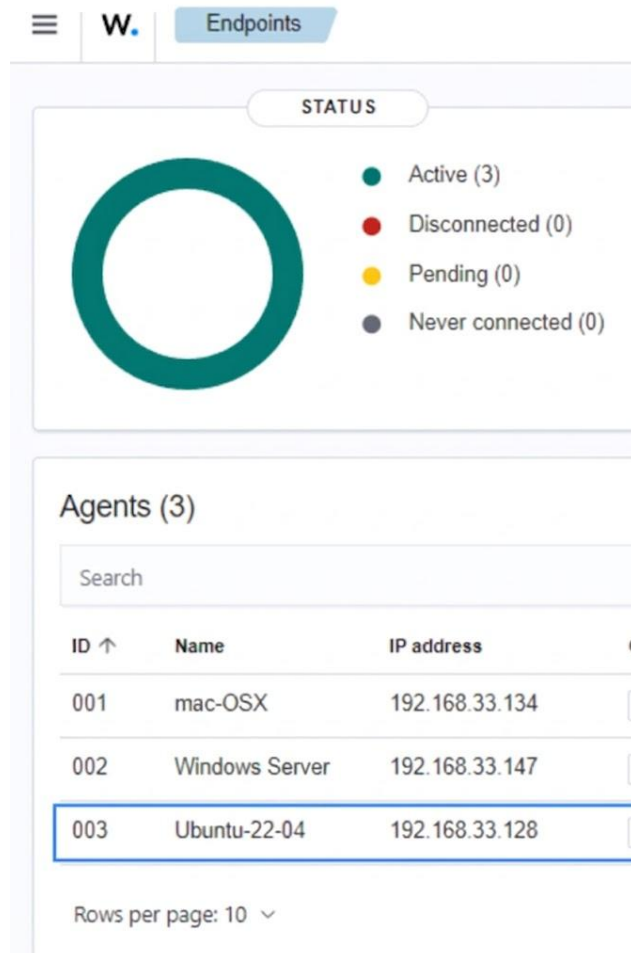


Fig.6 Wazuh Endpoints summary confirming three Active agents (Windows, Linux, and Mac).

Results & Dashboard Analysis

The implementation resulted in a high-fidelity monitoring stream. The Wazuh dashboard provided the following technical insights:

- **Network Visibility:** Captured real-time traffic data, specifically identifying high-volume communication on Port 1514, the standard port for Wazuh agent-to-manager log forwarding.

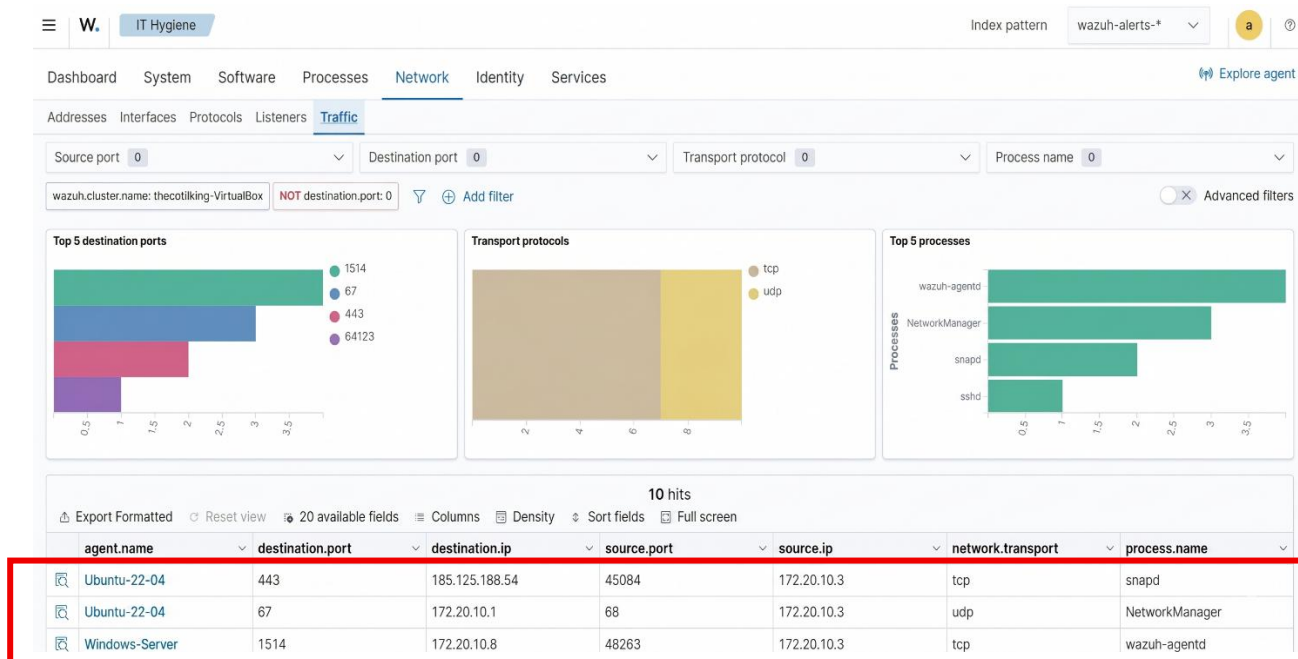


Fig.7 Network Traffic Analysis identifying log forwarding activity on Port 1514.

- **System Inventory:** Successfully gathered IT Hygiene data from the Windows-Server VM, including a detailed inventory of browser extensions such as Microsoft Edge and Internet Explorer.

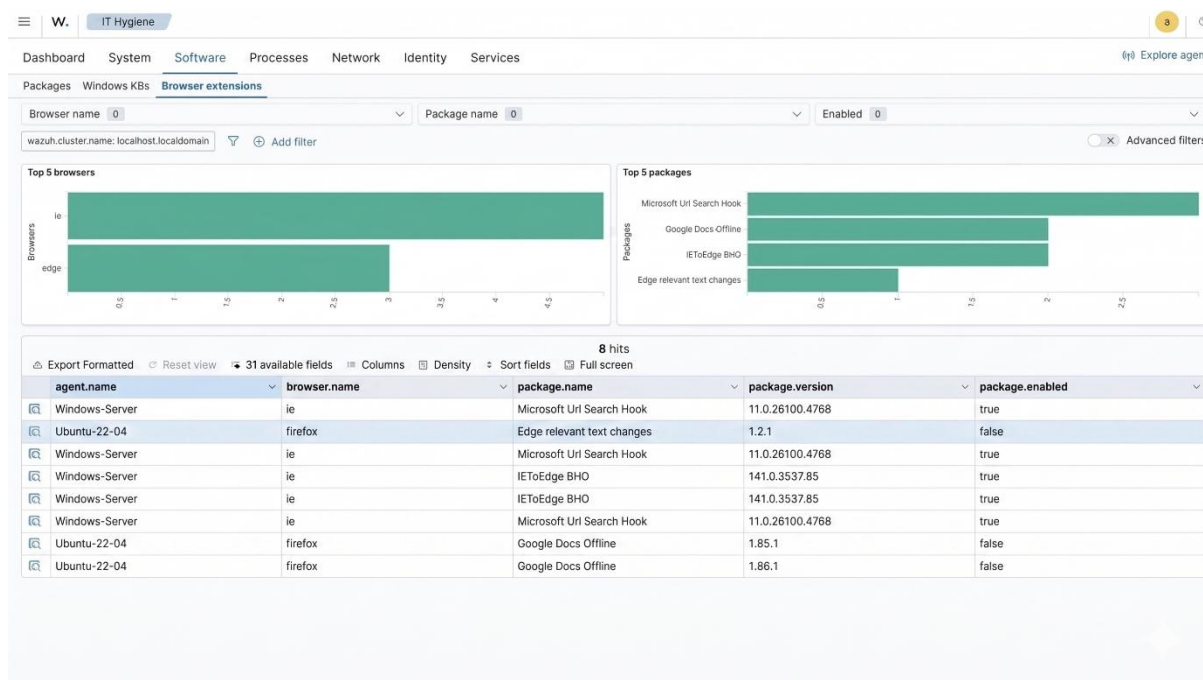


Fig.8 Software Inventory (IT Hygiene) dashboard displaying Windows browser and Ubuntu extensions.

- **Active Heartbeats:** The Evolution chart and agent list confirmed a 100% coverage rate across the deployed nodes.



Fig.9 Coverage rate across deployed nodes

Analysis & Troubleshooting

A critical part of the deployment involved resolving connectivity and configuration barriers:

- **API Resolution:** During the initial setup, a critical *API connection offline* error was encountered on the dashboard. This required troubleshooting the communication between the Wazuh web interface and the underlying manager API to restore synchronization.

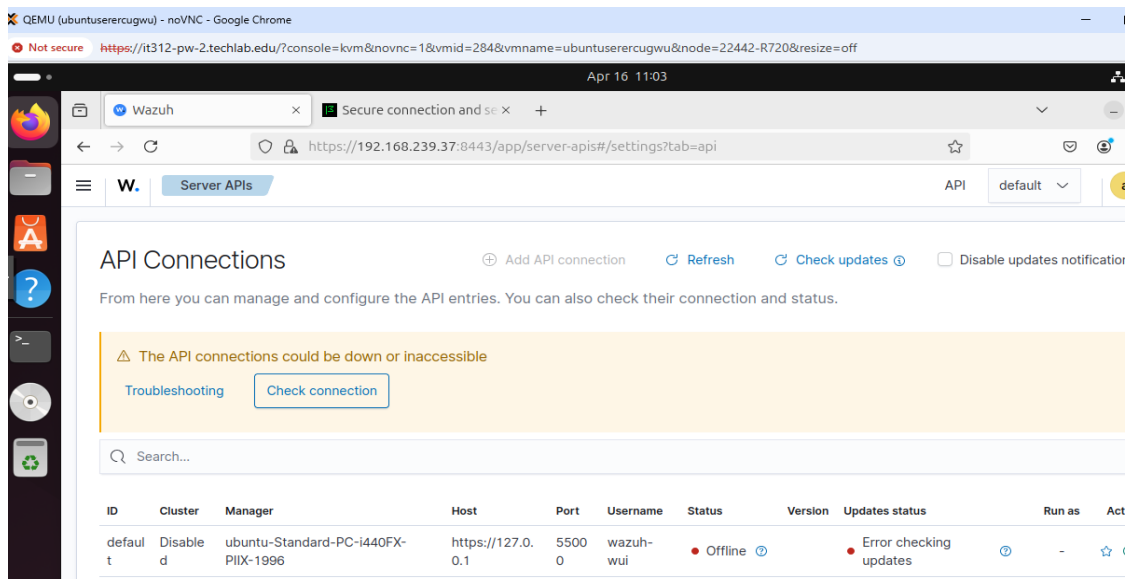


Fig.10 Wazuh Dashboard showing the resolution of the "API Connection Offline" error.

- **Attack Simulation Analysis:** A brute-force attack was launched from Kali Linux using the Metasploit `ssh_login` module.

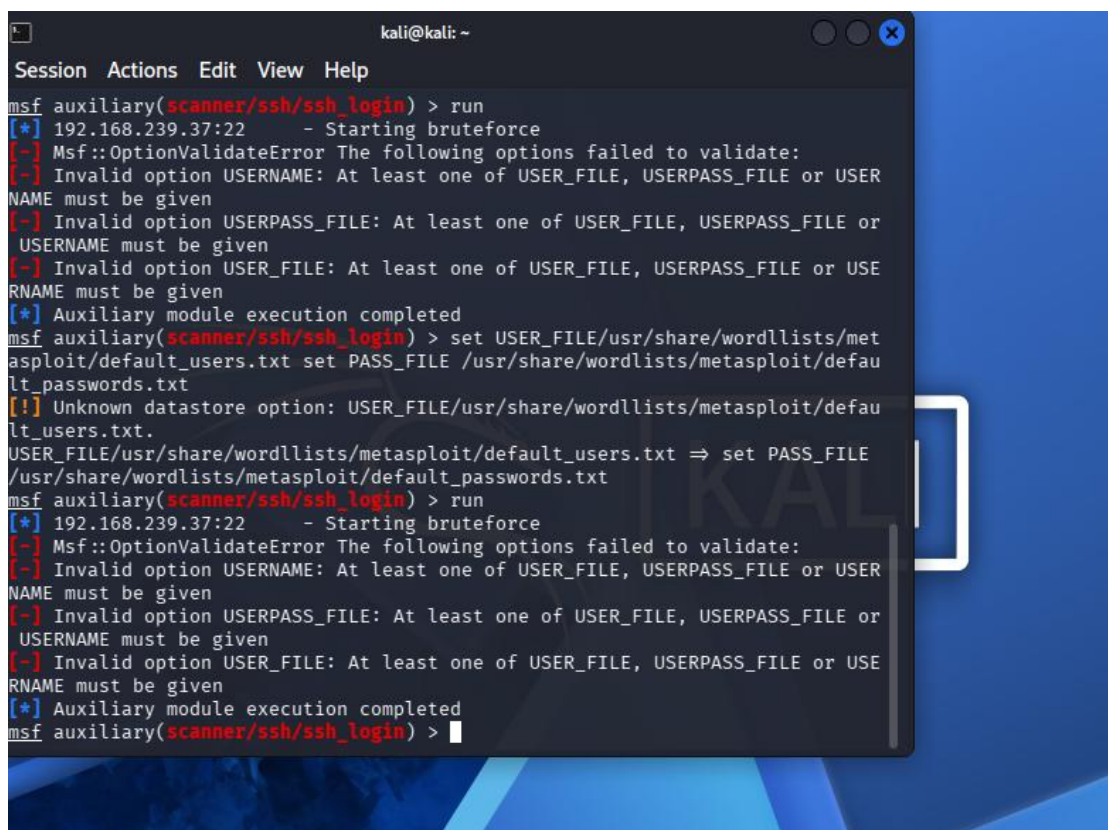


Fig.11 Kali Linux Metasploit terminal showing troubleshooting of the `ssh_login` module

- **Finding:** Initial attempts failed with an *OptionValidateError* due to missing wordlist paths.
- **Detection:** Once corrected, the system successfully tracked the inbound traffic, correlating the attack to the *sshd* process and source IP 172.20.10.3. (Please see Fig.7 above)

Recommendations

1. **Vulnerability Cross-Referencing:** Use the software inventory data (as seen in Fig.8) to automate the detection of CVEs in the browser extensions.
2. **Secure Communication:** Hardened firewall rules should be implemented to ensure only registered agent IPs can communicate over Port 1514.

Conclusion

This project successfully established a SOC pipeline within a Proxmox environment, integrating HIDS agents and validating their detection capabilities. The exercise provided practical experience in SIEM registration, API troubleshooting, and traffic analysis, fulfilling all requirements for an enterprise-ready IDS infrastructure.